PRAJJAWAL SINGH

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Application for Backend Engineer

Dear Hiring Manager,

I am applying for a Backend Engineer opportunity where I can contribute my production engineering experience and skills in Java 17+, Spring Boot, Python and SQL. I hold a B.Tech from NIT Tiruchirappalli and work as a Backend Engineer at TailNode, building APIs, data pipelines and cloud-hosted services.

One customer problem I addressed combined slow shape and coordinate loading with rising cloud costs. I investigated content delivery, coordinate payload sizes, S3 usage and Kafka replication traffic across Availability Zones to identify the main drivers.

I implemented CDN caching that reduced average load time from 5 seconds to 2 seconds (60%) and monthly origin data-transfer costs from $500 to $300 (40%). I also integrated a WAF to filter malicious requests and protect application endpoints. By removing redundant coordinates and optimising precision while preserving required shape accuracy, I reduced average payload size from 2 MB to 800 KB (60%), helping users download, render and interact with shapes more smoothly.

I optimised stored data and applied appropriate S3 lifecycle policies, reducing monthly storage costs from $300 to $210 (30%). I moved a suitable messaging workload from Kafka to WarpStream to reduce cross-Availability Zone replication traffic, lowering monthly messaging infrastructure costs from $700 to $420 (40%). Across these three categories, monthly spend fell from $1,500 to $930: a saving of $570, or 38%.

Beyond this work, I automated a Kafka-to-ClickHouse pipeline handling 500,000+ daily events, developed secured REST and FastAPI endpoints at sub-50ms p99 latency, and deployed services on Kubernetes using Helm. These experiences have taught me to connect technical decisions with customer experience, reliability and operating cost.

I would welcome the opportunity to bring this approach and my strong foundation in algorithms and backend system design to your team. Thank you for considering my application.

Sincerely,
Prajjawal Singh